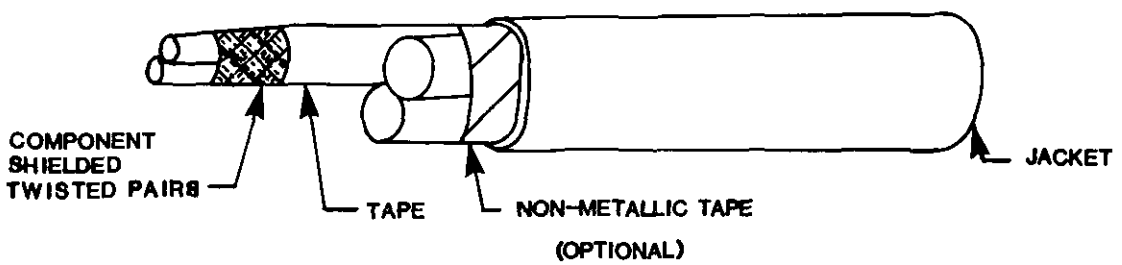
MIL-C-0024640/15A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/15
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, AWG 22, TYPE 2XSAW (INCLUDING VARIATION TYPES 2XSAOW AND 2XSAWA)

This military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24640. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24640.



SH 12762

COMPONENT SHIELDED PAIR:

First - Copper conductor, size AWG 22, 7 stranding. Nominal diameter of stranded conductor 0.030 inch.
Second - Material requirements in accordance with MIL-W-81044/12 except the conductor and identification codes and methods shall be in accordance with MIL-C-24640. Finished wire diameter 0.044 inch minimum and 0.050 inch maximum.
Third - Shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (watertight).

(A) denotes changes.

FSC 6145

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(A) TABLE I. Construction details - unarmored.

Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
2XSAW-3	3	0.041	0.368	0.396	112.0
2XSAW-7	7	.041	.461	.497	195.0
2XSAW-14	14	.044	.641	.691	384.0

(A) TABLE II. Construction details - armored.

Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
2XSAWA-3	3	0.041	0.418	0.446	147.8
2XSAWA-7	7	.041	.511	.547	243.8
2XSAWA-14	14	.041	.691	.741	445.4

(A) TABLE III. Construction details - overall shield.

Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inches)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
2XSAOW-3	3	0.291	0.041	0.405	0.437	153.0
2XSAOW-7	7	.400	.044	.510	.550	253.0
2XSAOW-10	10	.533	.044	.631	.681	362.0
2XSAOW-14	14	.585	.047	.689	.743	442.0
2XSAOW-19	19	.656	.047	.757	.817	545.0
2XSAOW-24	24	.791	.047	.884	.952	720.0
2XSAOW-30	30	.843	.050	.941	1.02	833.0
2XSAOW-37	37	.924	.050	1.01	1.09	971.0

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REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

When this cable is manufactured with armor, the following additional requirements shall apply:

(a) Braided metal armor and paint shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(a) An overall tin-coated copper braid shall be added.
(b) Braid coverage shall be not less than 85 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	17.03
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Conductor to shield	1000
Shield to shield (type 2XSAOW only)	500
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity (type 2XSAOW only)	No failure
Jacket flaws (type 2XSAOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	40
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	60 ± 5
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	3
Gas flame (1 hour)	Not applicable

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type 2XSAWA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance (type 2XSAOW only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-N322-15)

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